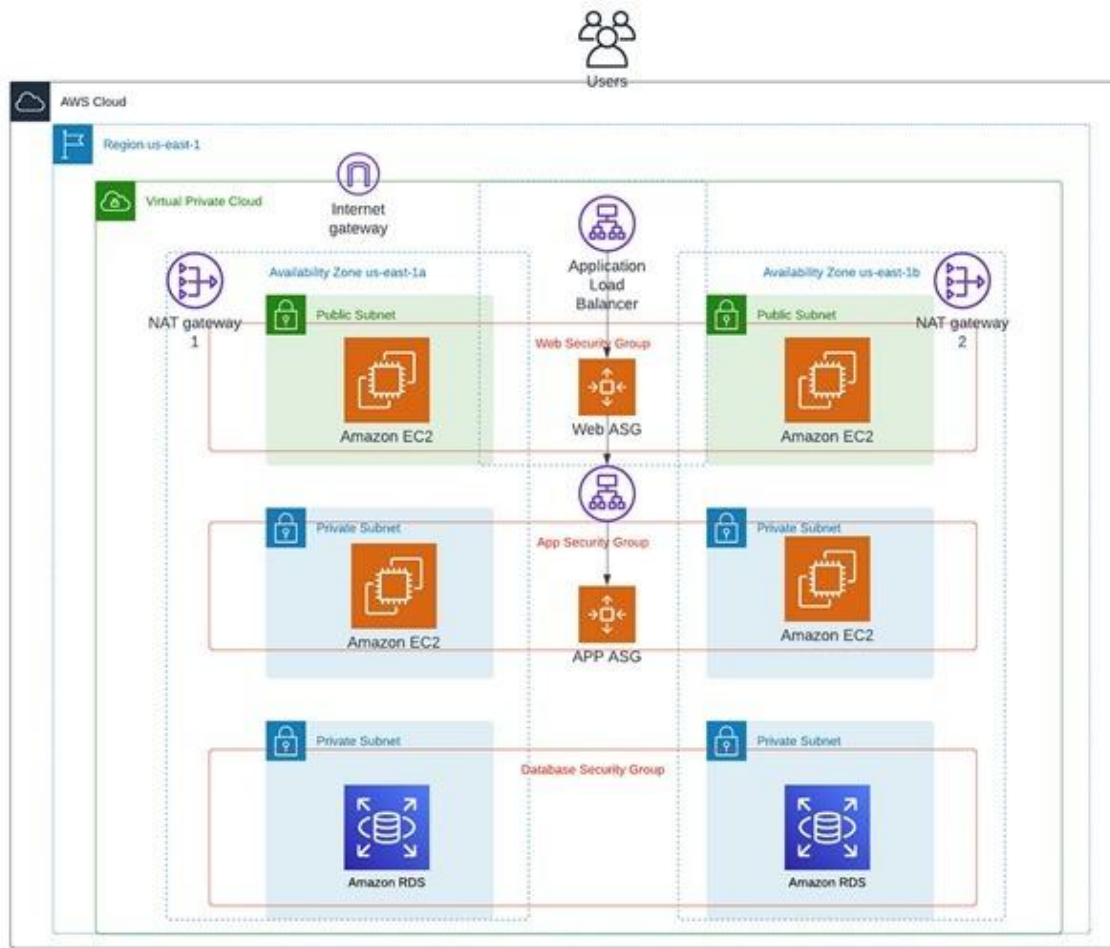


AWS Three-Tier Architecture Project Document

Project Architecture

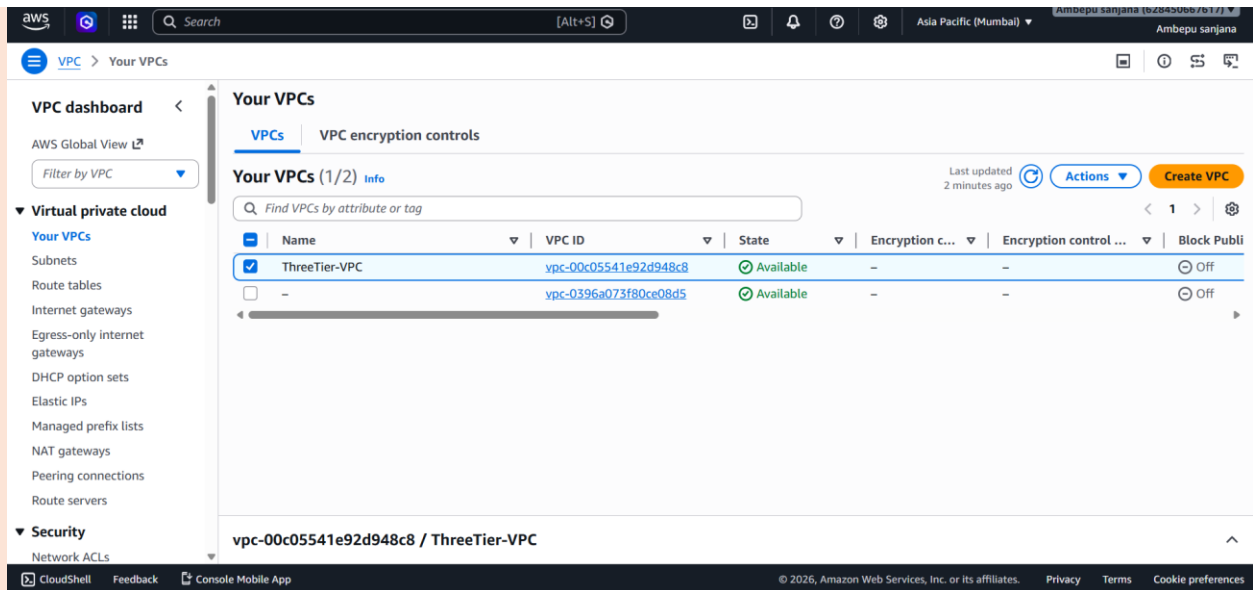
The project contains a three-tier architecture in AWS: Web Tier, Application Tier, and Database Tier. The design uses public subnets for the web layer and private subnets for the application and database layers across two Availability Zones for high availability.



Implementation Process and Screenshot Guide

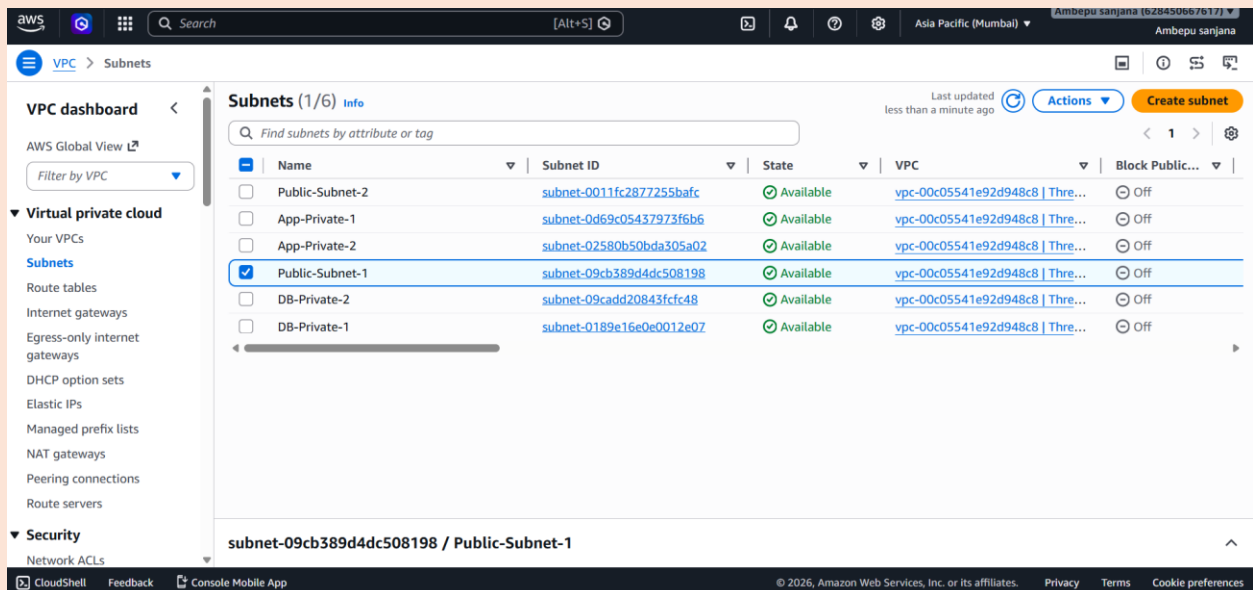
Step 1: Create VPC

1. Open AWS Console and go to VPC service.
2. Create a new VPC for the project with a suitable CIDR block, for example 10.0.0.0/16.
3. Name the VPC clearly, such as Three-Tier-VPC.
4. Enable DNS hostnames and DNS resolution if required.



Step 2: Create Subnets in Two Availability Zones

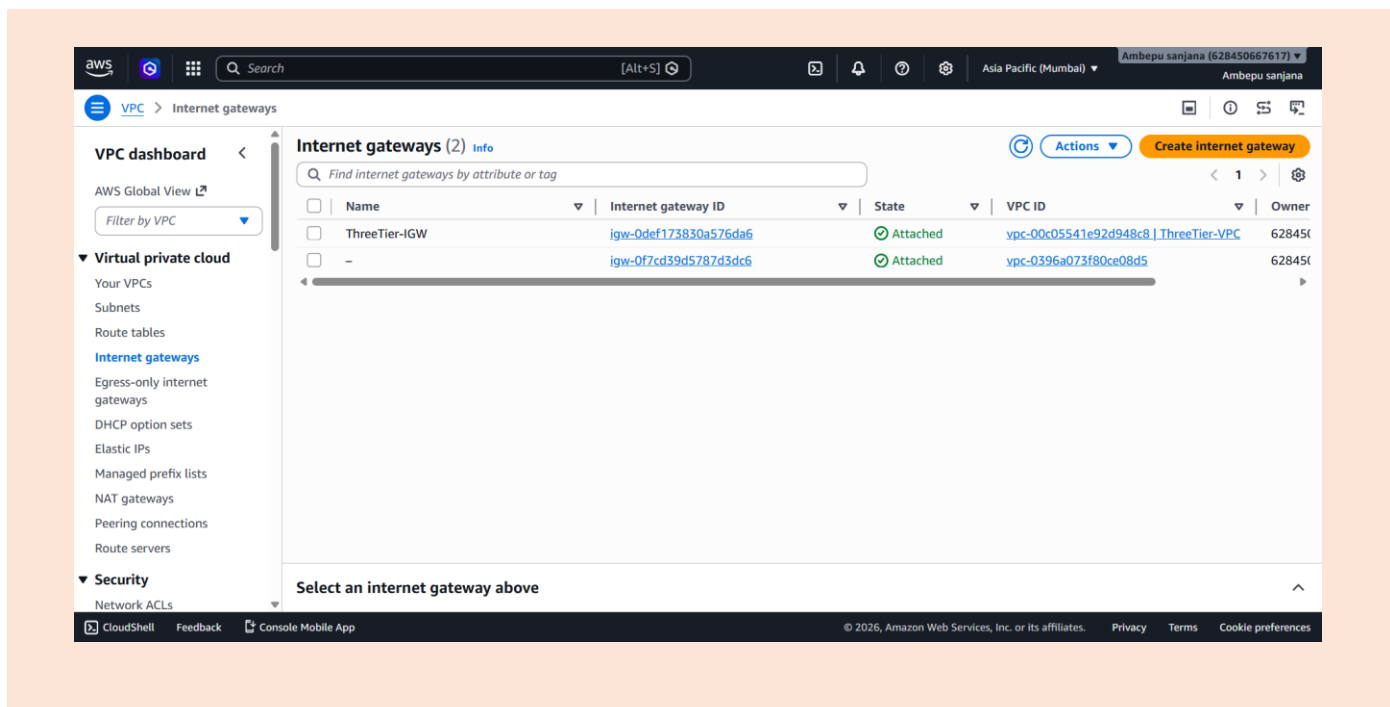
5. Create two public subnets, one in each Availability Zone.
6. Create two private subnets for the application tier, one in each Availability Zone.
7. Create two private subnets for the database tier, one in each Availability Zone.
8. Use proper names like Public-Subnet-AZ1, App-Private-Subnet-AZ1, DB-Private-Subnet-AZ1.



Step 3: Attach Internet Gateway

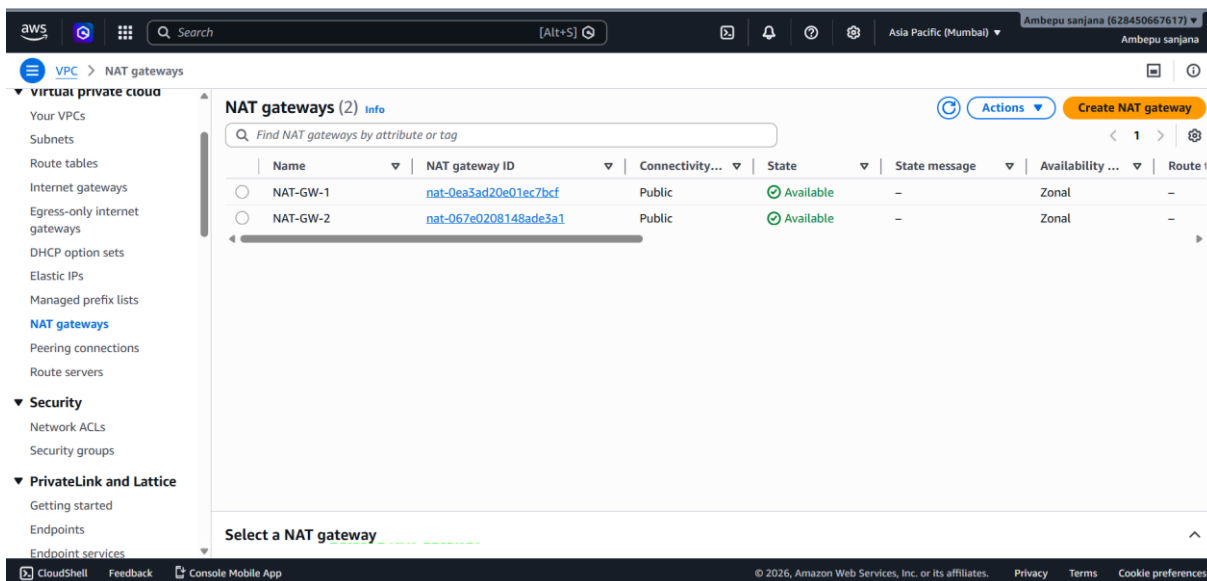
9. Go to Internet Gateways and create a new Internet Gateway.

10. Attach it to the project VPC.
11. This allows internet access for resources in public subnets through route tables.



Step 4: Create NAT Gateways

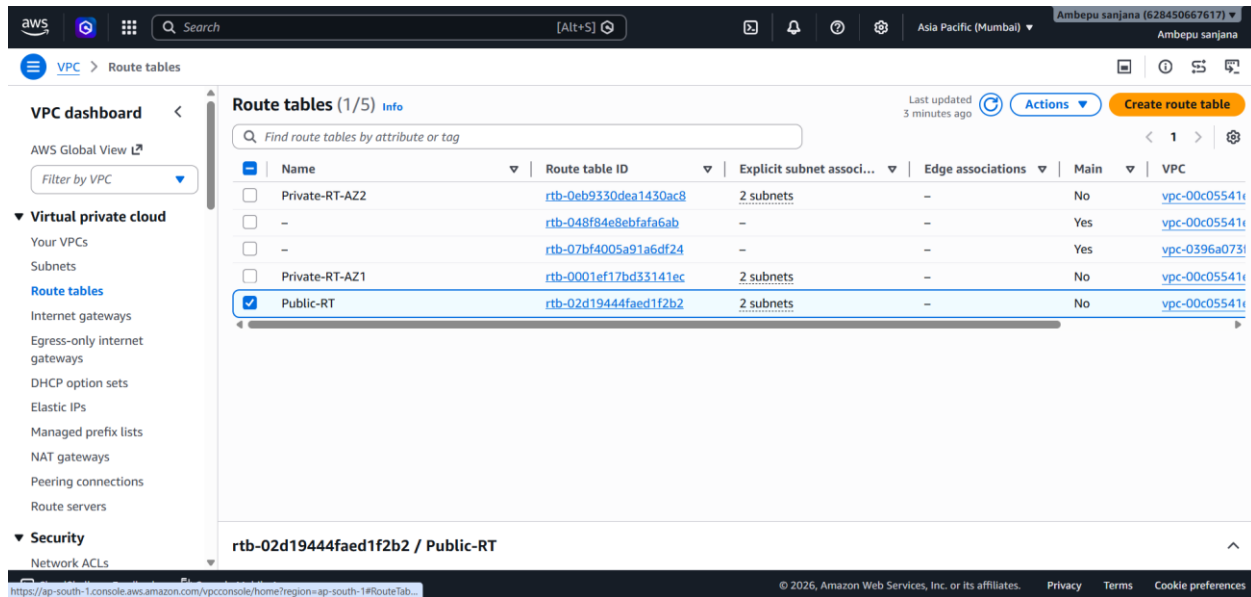
12. Create NAT Gateway 1 in Public Subnet AZ1 with an Elastic IP.
13. Create NAT Gateway 2 in Public Subnet AZ2 with another Elastic IP.
14. NAT Gateway allows private EC2 instances to access the internet for updates without allowing inbound internet traffic



Step 5: Configure Route Tables

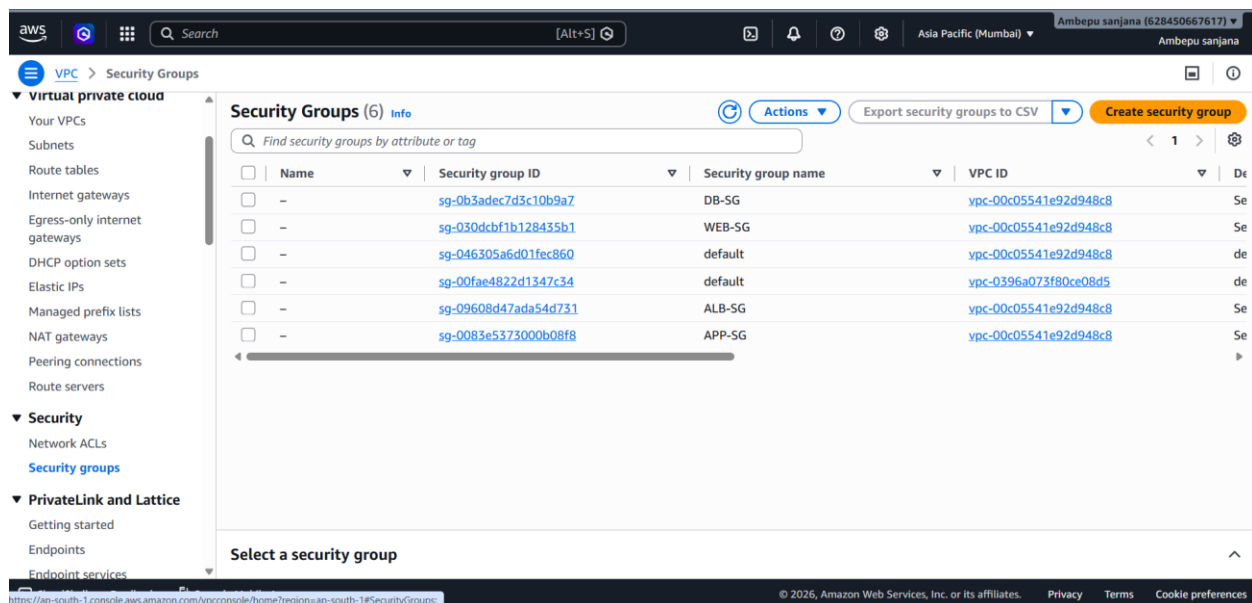
15. Create a public route table and associate it with both public subnets.

16. Add route 0.0.0.0/0 pointing to the Internet Gateway.
17. Create private route table for AZ1 and add route 0.0.0.0/0 pointing to NAT Gateway 1.
18. Create private route table for AZ2 and add route 0.0.0.0/0 pointing to NAT Gateway 2.
19. Associate app and database private subnets with their correct private route tables.



Step 6: Create Security Groups

20. Create Web Security Group: allow HTTP/HTTPS from internet and required health check traffic.
21. Create App Security Group: allow traffic only from Web Security Group on application port.
22. Create Database Security Group: allow MySQL port 3306 only from App Security Group.
23. Do not open database access to 0.0.0.0/0.



Step 7: Create Web EC2 Launch Template

24. Create a launch template for web servers.

25. Select AMI, instance type, key pair if needed, and Web Security Group.
26. Add user data if required to install web server packages.
27. Save the launch template.

The screenshot shows the AWS Management Console interface for the 'Launch Templates' section. The left sidebar contains navigation links for EC2, Instances, Images, and Elastic Block Store. The main content area displays a table of launch templates. The 'Web-LT' template is selected, and its details are shown below the table.

Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By
lt-08ae51bcab42fc933	App-LT	1	2	2026-06-26T16:43:53.000Z	arn:aws:iam::628450667617:root
lt-0731a4f4f09c764b2	Web-LT	1	3	2026-06-26T16:31:21.000Z	arn:aws:iam::628450667617:root

Web-LT (lt-0731a4f4f09c764b2)

Launch template details

Launch template ID	Launch template name	Default version	Owner
lt-0731a4f4f09c764b2	Web-LT	1	arn:aws:iam::628450667617:root

Launch template version details

Step 8: Create Web Auto Scaling Group

28. Create Web-ASG using the Web Launch Template.
29. Select public subnets in both Availability Zones.
30. Attach the Web-ASG to the Application Load Balancer target group.
31. Set desired capacity, minimum capacity, and maximum capacity based on trainer requirements.

The screenshot shows the AWS Management Console interface for the 'Auto Scaling groups' section. The left sidebar contains navigation links for Volumes, Snapshots, Lifecycle Manager, Network & Security, Load Balancing, Auto Scaling, and Settings. The main content area displays a table of auto scaling groups. The 'Web-ASG' group is shown with 2 instances.

Name	Status - new	Launch template/configuration	Instances	Instance health - new
App-ASG	At desired capacity	App-LT Version Default	1	1/1 Healthy
Web-ASG	At desired capacity	Web-LT Version Default	2	2/2 Healthy

0 Auto Scaling groups selected

Step 9: Create Application Load Balancer

32. Create an internet-facing Application Load Balancer.
33. Select the project VPC and both public subnets.
34. Attach the Web Security Group or ALB security group as configured.
35. Create listener on HTTP port 80 and forward traffic to the web target group.

The screenshot shows the AWS Management Console for the 'Load balancers' section. On the left, the navigation pane includes 'Volumes', 'Snapshots', 'Lifecycle Manager', 'Network & Security', 'Load Balancing', and 'Auto Scaling'. The 'Load balancers' page displays a table with one entry: 'Web-ALB' in an 'Active' state, of type 'application' and 'Internet-facing', using 'IPv4' IP address type and VPC 'vpc-00c05541e92d948c8'. Below the table, the 'Details' tab for 'Web-ALB' is active, showing fields for 'Load balancer type' (Application), 'Status' (Active), 'VPC' (vpc-00c05541e92d948c8), 'Load balancer IP address type' (IPv4), 'Scheme', 'Hosted zone', 'Availability Zones', and 'Date created'.

Step 10: Create App EC2 Launch Template

36. Create a launch template for app servers.
37. Select AMI, instance type, and App Security Group.
38. Place app servers in private subnets using the Auto Scaling Group.
39. Add backend application installation commands in user data if required.

The screenshot shows the AWS Management Console for the 'Launch Templates' section. The left navigation pane includes 'EC2', 'Instances', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', 'Images', and 'Elastic Block Store'. The 'Launch Templates' page displays a table with two entries: 'App-LT' and 'Web-LT'. The 'Web-LT' is selected, and its details are shown in the 'Web-LT (lt-0731a4f4f09c764b2)' section. The 'Launch template details' show the 'Launch template ID' as 'lt-0731a4f4f09c764b2', 'Launch template name' as 'Web-LT', 'Default version' as '1', and 'Owner' as 'arn:aws:iam::628450667617:root'. The 'Details' tab is active, showing 'Versions' and 'Template tags'.

Step 11: Create App Auto Scaling Group

40. Create App-ASG using App Launch Template.
41. Select the application private subnets in both Availability Zones.
42. Set desired, minimum, and maximum capacity.
43. Confirm instances are launched in private subnets.

The screenshot shows the AWS Management Console interface for 'Auto Scaling groups'. The left-hand navigation pane is expanded, showing categories like 'Network & Security', 'Load Balancing', and 'Auto Scaling', with 'Auto Scaling Groups' selected. The main content area is titled 'Auto Scaling groups (2)' and includes buttons for 'Launch configurations', 'Launch templates', 'Actions', and 'Create Auto Scaling group'. Below this is a search bar and a table listing the existing groups.

☐	Name	Status - new	Launch template/configuration	Instances	Instance health - new
☐	App-ASG	✔ At desired capacity	App-LT Version Default	1	✔ 1/1 Healthy
☐	Web-ASG	✔ At desired capacity	Web-LT Version Default	2	✔ 2/2 Healthy

At the bottom of the console, a status bar indicates '0 Auto Scaling groups selected'.

Step 12: Create DB Subnet Group

44. Open RDS service and go to Subnet Groups.
45. Create a DB subnet group using the two database private subnets.
46. Select the correct VPC and Availability Zones.

Aurora and RDS > Subnet groups

Subnet groups (1)

Filter by subnet group

☐	Name	Description	Status	VPC
☐	projectsubnetgroup	allow	Complete	vpc-00c05541e92d948c8

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Step 13: Create RDS Database

47. Create an Amazon RDS MySQL database.
48. Select DB subnet group created in the previous step.
49. Keep public access disabled.
50. Attach the Database Security Group.
51. Create single-AZ or multi-AZ based on trainer requirement.

Aurora and RDS > Databases

Databases (1)

Group resources

Filter by databases

☐	DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Si
☒	projectdatabase-1	Available	Instance	MySQL Co...	SECOND	ap-south-1b	db

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Step 14: Connect App Server to RDS

52. Connect to the app EC2 instance using SSM Session Manager or bastion host.

53. Install MySQL client if not already installed.
54. Use the RDS endpoint, username, and password to connect.
55. Test database connection from app EC2 to RDS.

```
aws  [Search] [Alt+S]  Asia Pacific (Mumbai)  Ambepu sanjana (628450667617)  Ambepu sanjana

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sat Jun 27 08:42:13 2026 from 13.233.177.3
[ec2-user@ip-10-0-1-147 ~]$ sudo dnf update -y
Last metadata expiration check: 2:44:26 ago on Sat Jun 27 06:53:07 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-1-147 ~]$ sudo dnf install mariadb105 -y
Last metadata expiration check: 2:44:32 ago on Sat Jun 27 06:53:07 2026.
Package mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-1-147 ~]$ mysql --version
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper
[ec2-user@ip-10-0-1-147 ~]$ mysql -h projectdatabase-1.cvmsoeqobnp.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
```

i-034bf0da9f3217d5f (WebServer1)
PublicIPs: 52.66.214.186 PrivateIPs: 10.0.1.147

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```
aws  [Search] [Alt+S]  Asia Pacific (Mumbai)  Ambepu sanjana (628450667617)  Ambepu sanjana

information_schema |
mysql              |
performance_schema |
projectdb          |
sys                |
+-----+
5 rows in set (0.010 sec)

MySQL [(none)]> USE projectdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [projectdb]> CREATE TABLE employees (
  -> CREATE TABLE employees ( id INT PRIMARY KEY AUTO INCREMENT, name VARCHAR(50), department VARCHAR(50) );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'C
REATE TABLE employees ( id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR' at line 2
MySQL [projectdb]> SELECT * FROM employees;
+----+-----+-----+
| id | name  | department |
+----+-----+-----+
| 1  | Sanjana | DevOps      |
+----+-----+-----+
1 row in set (0.006 sec)

EXIT;
Bye
[ec2-user@ip-10-0-1-147 ~]$
```

i-034bf0da9f3217d5f (WebServer1)
PublicIPs: 52.66.214.186 PrivateIPs: 10.0.1.147

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